

CAMERA CONTROLLER – signal flow

GND is a global net on every sheet. Double-click a block to open it.

1. Battery 2S + protection



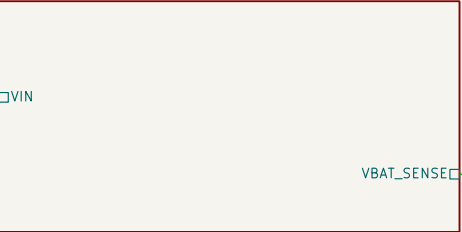
File: sheets/power_in.kicad_sch

2. Buck to 5 V



File: sheets/buck_5v.kicad_sch

3. Pack voltage sense



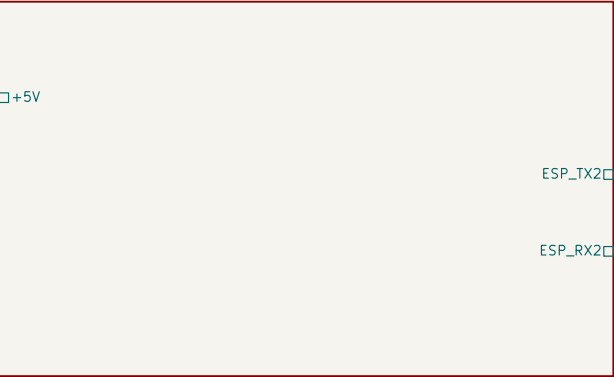
File: sheets/sense.kicad_sch

4. ESP32 socket



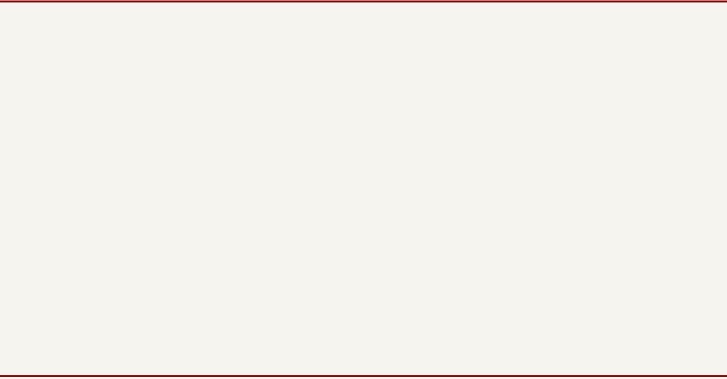
File: sheets/esp32.kicad_sch

5. RunCam Split 4



File: sheets/camera.kicad_sch

6. COTS / SRAD supplies

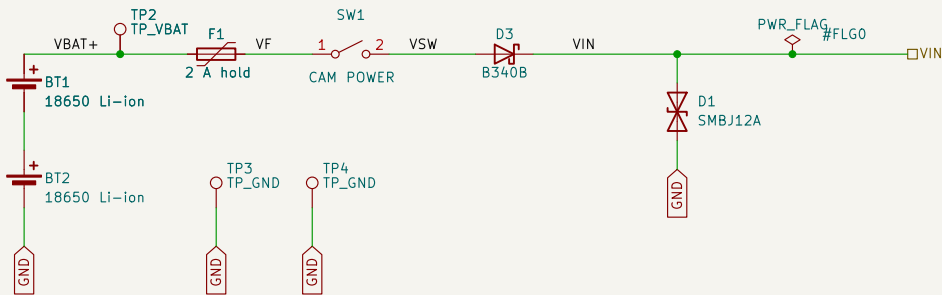


File: sheets/supplies.kicad_sch

Block 6 is electrically isolated from the rest of this board on purpose.

2 x 18650 in series: 7.4 V nominal, 8.4 V full. Charge the cells off the board, one at a time, in a Li-ion charger.

F1: 2 A PTC right at the pack +. D3: reverse-polarity Schottky (B340B, about 0.35 V at the 0.6 A load).
D1: TVS, 12 V standoff (above the 8.4 V full pack), clamps below 20 V; everything on VIN is rated 25 V or more.
Low-battery cut-off: U1's EN divider on sheet 2 (about 2.9 V per cell).



Stanford SSI

Sheet: /1. Battery 2S + protection/
File: power_in.kicad_sch

Title: 1. Camera + ESP32 supply (2S)

Size: A4
KiCad E.D.A. 10.0.0

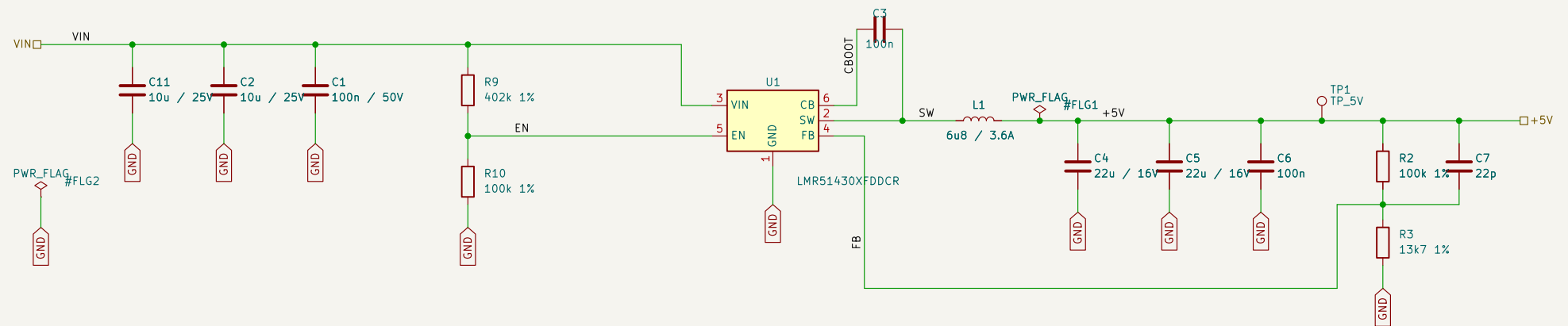
Date:

Rev: A
Id: 2/7

LMR51430 buck, 500 kHz: $V_{OUT} = 0.6 \times (1 + 100k / 13k7) = 4.98 \text{ V}$

Input caps: C1 (100 nF) sits right at U1's VIN/GND pins; C2, C11 (10 uF) are the bulk.

UVLO (R9/R10 on EN): on above $1.23 \text{ V} \times (1 + 402k/100k) = 6.2 \text{ V}$, off below $1.08 \text{ V} \times 5.02 = 5.4 \text{ V}$ at VIN,
so the camera pack stops at about 2.85 V per cell instead of being run flat (U1's own UVLO is 3.6 V).



Stanford SSI

Sheet: /2. Buck to 5 V/

File: buck_5v.kicad_sch

Title: 2. Buck 8.4 V -> 5.0 V

Size: A4

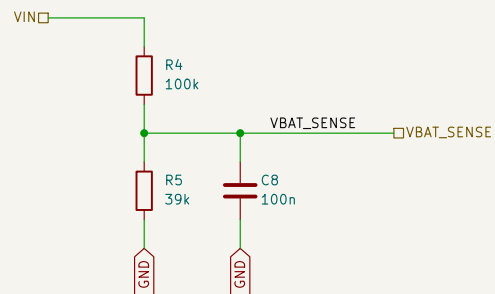
Date:

KiCad E.D.A. 10.0.0

Rev: A

Id: 3/7

$V_{BAT_SENSE} = V_{IN} \times 39k / (100k + 39k)$. Full pack 8.4 V (VIN about 8.1 V after D3) -> 2.26 V at GPIO34,
inside the ESP32 ADC's 2.45 V range (11 dB). Firmware: $V_{IN} = V_{BAT_SENSE} \times 139 / 39$; pack = VIN + about 0.35 V.



Stanford SSI

Sheet: /3. Pack voltage sense/

File: sense.kicad_sch

Title: 3. Pack voltage sense

Size: A4

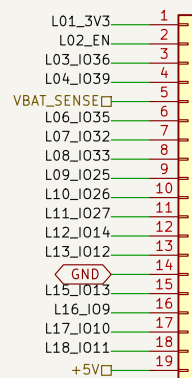
Date:

Rev: A

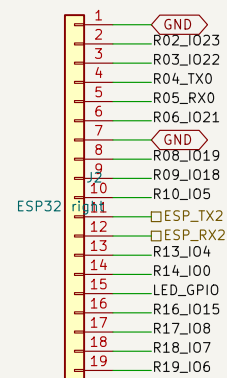
KiCad E.D.A. 10.0.0

Id: 4/7

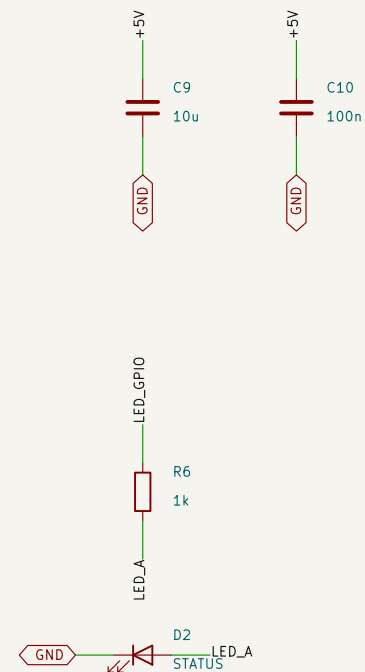
2 x 1x19 female header. GPIO6–11 go to the module SPI flash – do not use.



J1
ESP32 left



ESP32 right



Stanford SSI

Sheet: /4. ESP32 socket/

File: esp32.kicad_sch

Title: 4. ESP32–DevKitC V4 socket

Size: A4

Date:

Rev: A

KiCad E.D.A. 10.0.0

Id: 5/7

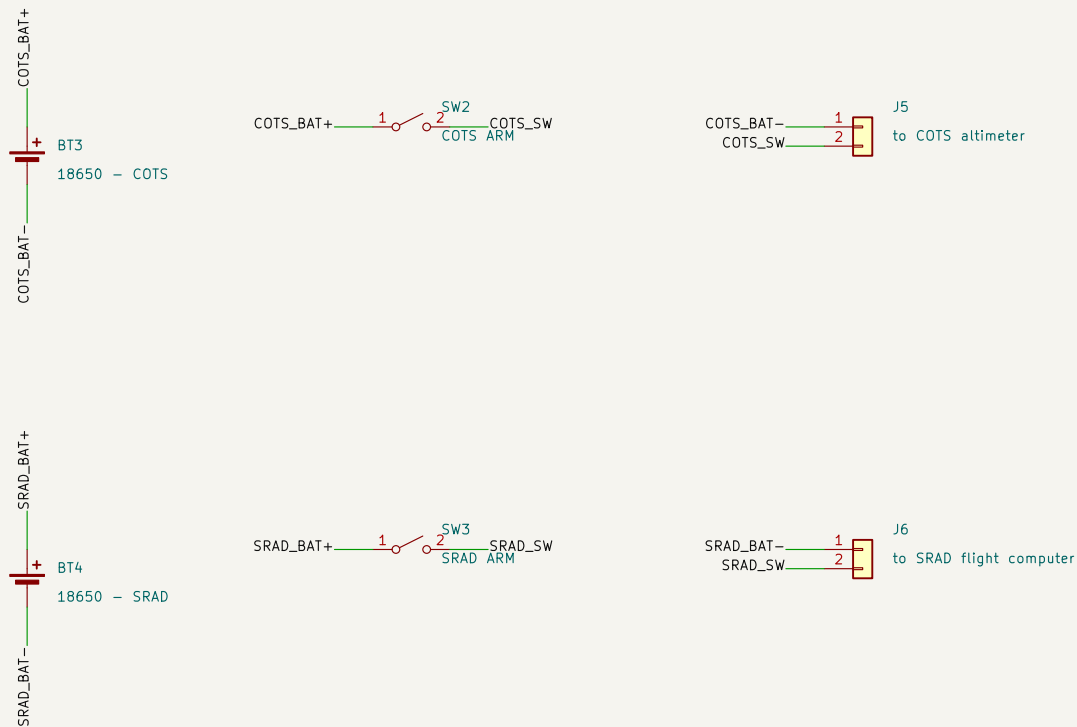
5–20 V in. Single 5–pin harness: power, ground and UART.



Stanford SSI		
Sheet: /5. RunCam Split 4/		
File: camera.kicad_sch		
Title: 5. RunCam Split 4		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0	Id: 6/7	

Own cell, own switch, per DTEG 6.9.1. SW2 / SW3 take the pin-switch leads; they sit next to the pin-switch stack. All connectors latch (Micro-Fit 3.0).

These returns are deliberately NOT tied to GND – that independence is the point.



Stanford SSI		
Sheet: /6. COTS / SRAD supplies/ File: supplies.kicad_sch		
Title: 6. Independent 1S supplies		
Size: A4	Date:	Rev: A
KiCad E.D.A. 10.0.0	Id: 7/7	